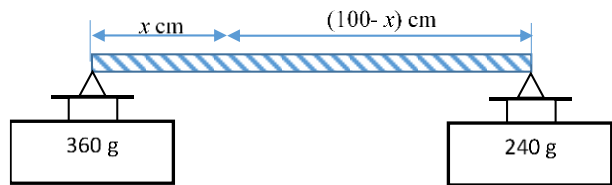


Q5.



Let the centre of gravity be x in cm from the left hand end of the rod. Then the c.g. will be $(100 - x)$ in cm from the right hand end of the rod. Taking moments about the c.g. should give net zero moments. Hence

$$(360 \times 10^{-3})g x = (240 \times 10^{-3})g (100 - x)$$

$$600x = 24000$$

$$x = 40 \text{ cm}$$

As the geometrical centre is at the 50 cm mark, the c.g. is 10 cm left of the geometrical centre.

Ans: A